Aryan Manoj Jadhav

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Summary

Final-year B.Tech Computer Science and Engineering student at LNMIIT working in Human-Computer Interaction. Research experience in behavioural study design and psychometric scale development, studying how UI patterns shape user cognition and decision-making. Holds the Google UX Design Professional Certificate and has industry exposure through a UX research internship, with hands-on experience across the design process from user research and wireframing to high-fidelity prototyping in Figma and usability testing. Brings an engineering background in Python, Java, and deep learning, with a research orientation toward interpretable, trustworthy, and human-centered systems.

Research Interests: Human-Computer Interaction, Digital Nudging & Choice Architecture, Behavioural Research Methods, Human-Centered & Explainable AI

Education

The LNM Institute of Information Technology (LNMIIT), Jaipur *Aug 2023 – May 2027 (Expected)*
B.Tech in Computer Science and Engineering CGPA: 6.88/10 (through Semester 6)
Relevant Coursework: Operating Systems, Database Management Systems, Computer Networks, Computer Architecture

Springdale Jr. College, Aurangabad — Class XII (CBSE): 77.83% 2023

Nath Valley School, Aurangabad — Class X: 92.4% 2020

Experience

Enteledge Private Limited *May 2026 – Jul 2026*
Intern, UX Research & Workflow Engineering *Aurangabad, India*
Project: Improving Asynchronous Client Communication in Distributed BIM Projects

- Conducted a user-centered study of how India-based Building Information Modeling (BIM) teams communicate with overseas clients who have little or no overlap in working hours, a handoff process that had never been formally studied or designed.
- Examined the clarification handoff between modelers and clients, where questions about missing drawings, ambiguous scope items, or design conflicts often waited a full working day for a reply, and unclear answers added further cycles, slowing delivery, causing rework, and straining client relationships.
- Designed a **Figma prototype** of a redesigned clarification handoff, aimed at getting clear, complete answers from clients in a single cycle.

Tools: Figma, User Research

Projects

Digital Nudging & User Behaviour *Under Dr. Rajbala Singh | Ongoing*

- Investigating how UI design elements influence user cognition and decision-making.
- Developed and validated the **Perceived Digital Nudging Scale (PDNS)**, a psychometric instrument classifying digital nudges across Self-Determination Theory and compliance principles.
- Next phase: testing a moderated mediation model linking nudge type, cognitive effort, impulsivity, and decision confidence.

Tools: Python, SPSS, R, Figma, Behavioural Research Methods

Public Museum App *Google UX Design Capstone Project*

- Designed a mobile app for a public museum end to end, from user research to an interactive high-fidelity prototype.
- Conducted user research and built personas, empathy maps, and user journey maps to define visitors' pain points.
- Created wireframes and low-fidelity prototypes, ran a usability study, and iterated on the findings to produce high-fidelity mockups in Figma.

Tools: Figma, User Research, Wireframing, Prototyping, Usability Testing

Explainable AI-Based Threat Modeling for Cyber-Physical Systems *Under Dr. Ashish Kumar Dwivedi*

- Applied STRIDE threat modeling to the SWaT industrial control dataset (1.4M+ samples, 36 attack scenarios).
- Built an LSTM Autoencoder with Attention for unsupervised anomaly detection, achieving **96% accuracy**, **86% recall**, and **AUC = 0.997**.
- Produced STRIDE-linked, analyst-interpretable threat explanations using feature-level reconstruction error and sensor-type rule matching.
- Evaluated robustness against PGD adversarial attacks and hardened the model with weighted adversarial training (0.8/0.2 loss).

Tools: Python, PyTorch, LSTM Autoencoder, XGBoost, Scikit-learn, Pandas

IndiVision-Festival: Cultural Vision Dataset & ViT Ablation

- Co-built a 2,500-image Indian festival dataset to address Western-centric bias in vision benchmarks.
- Fine-tuned ViT-Base (**87.8% accuracy**) and ran a systematic ablation across all 144 attention heads, finding only 7.6% are high-importance, all concentrated in early layers.

Tools: Python, PyTorch, ViT (timm), Stable Diffusion, LoRA

Other Projects

Predictive Maintenance System

- Designed the UI/UX in Figma, built the real-time dashboard frontend, and led system architecture for an ML-based industrial failure prediction platform.

Tools: React, Figma, Scikit-learn, MySQL

Techno-Economic Analysis of India's National Green Hydrogen Mission *Under Dr. Somnath Biswas*

- Conducted a sensitivity analysis of the Mission's $1/kg target, modeling the levelised cost of hydrogen (LCOH) across 32 parameter combinations using IEA/IRENA methodology.

Fake News Detector

- Full-stack web app combining OWL ontology reasoning with ML classification; developed the ontology and contributed across data preprocessing, model training, backend, and frontend.

Tools: Python, OWL, Flask, HTML/CSS/JavaScript

Handwritten Digit Recogniser

- ML pipeline achieving **97.8% accuracy** using neural networks with ReLU/Softmax, with t-SNE visualization and performance analysis.

Tools: Python, Keras

2D Tile-Based Adventure Game

- Java game with a Swing GUI, pixel-perfect collision detection, 8-directional sprite animation, a 50×50 tile world with scrolling camera, and a 60 FPS game loop.

Tools: Java, Swing, Graphics2D

Certifications

Google UX Design Professional Certificate — Coursera *Sep 2026*

Credential ID: K2O5TZH79NKF | coursera.org/verify/professional-cert/K2O5TZH79NKF

Eight-course program covering the end-to-end UX design process:

- Foundations of User Experience (UX) Design
- Start the UX Design Process: Empathize, Define, and Ideate
- Build Wireframes and Low-Fidelity Prototypes
- Conduct UX Research and Test Early Concepts
- Create High-Fidelity Designs and Prototypes in Figma
- Build Dynamic User Interfaces (UI) for Websites
- Design a User Experience for Social Good & Prepare for Jobs
- Accelerate Your Job Search with AI

BlockBase 2025 — Finance & Economics Club, IIT Guwahati (DeFi & Blockchain) *Apr 2025*

Technical Skills

HCI & UX Design	Figma, User Research, Usability Testing, Wireframing, Low- & High-Fidelity Prototyping, UI Design
Research Methods	Psychometric Scale Development, Behavioural Study Design, SPSS, R
Programming	Python, Java, C, JavaScript
ML & Deep Learning	PyTorch, Keras, Scikit-learn, Hugging Face, Transformers, ViT, LSTM Autoencoders, XGBoost
Computer Vision	Semantic Segmentation, OpenCV, Attention Mechanisms
Web & Databases	React, Flask, HTML, CSS, REST APIs, MySQL, MongoDB
Tools	Git, OWL Ontology, Stable Diffusion, LoRA

Achievements & Leadership

- Secured **96.87 percentile** in JEE Main 2023 (AIR **36,025**).
- Content Creator, **TEDxLNMIIT** (2 editions).
- Head Organiser, **Ganesh Chaturthi @ LNMIIT**.
- Member, Media Cell & Literary Committee, LNMIIT.